

AI-ENHANCED ONLINE MULTIPLAYER GAME SERVER WITH PREDICTIVE OPPONENT MODELLING VIA WEBSOCKETS

TEOH WEI HONG

DR MOHAMMAD BABRDEL BONAB



PROBLEM STATEMENT

While gaming technology has rapidly advanced visual fidelity, gameplay design has not evolved at the same pace.

- Rigid Design:** Heavy reliance on hardcoded logic limits flexibility and increases development time
- Repetitive Gameplay:** City-building games lose engagement and replayability in later stages
- Lack of Guidance:** Performance metrics lack explanation, leaving players unsure how to improve

INTRODUCTION

AI's flexibility enables dynamic gameplay experiences that are less constrained by rigid, hardcoded outcomes. It allows systems to adapt to player behavior, creating more responsive and evolving interactions throughout gameplay.

Advantages of AI Pipeline:

- Reduced Development Overhead:** Minimizes reliance on extensive hardcoding
- Dynamic Gameplay:** Adapts to player actions and reduces repetitiveness
- Enhanced Immersion:** Delivers personalized feedback and tailored experience

OBJECTIVES

- Design a city-building game with strategic depth and varied playstyles
- Develop an AI pipeline for player modeling and adaptive gameplay
- Build a scalable server for concurrent single-player sessions
- Promote smart city concepts and United Nations Sustainable Development Goals awareness

GAME DESIGN

The game is a **turn-based city-building simulation** where players develop and manage a growing city. It features multiple building types, including residential, industrial, service, and infrastructure, each with distinct roles and effects.



DIFFERENT KINDS OF BUILDINGS

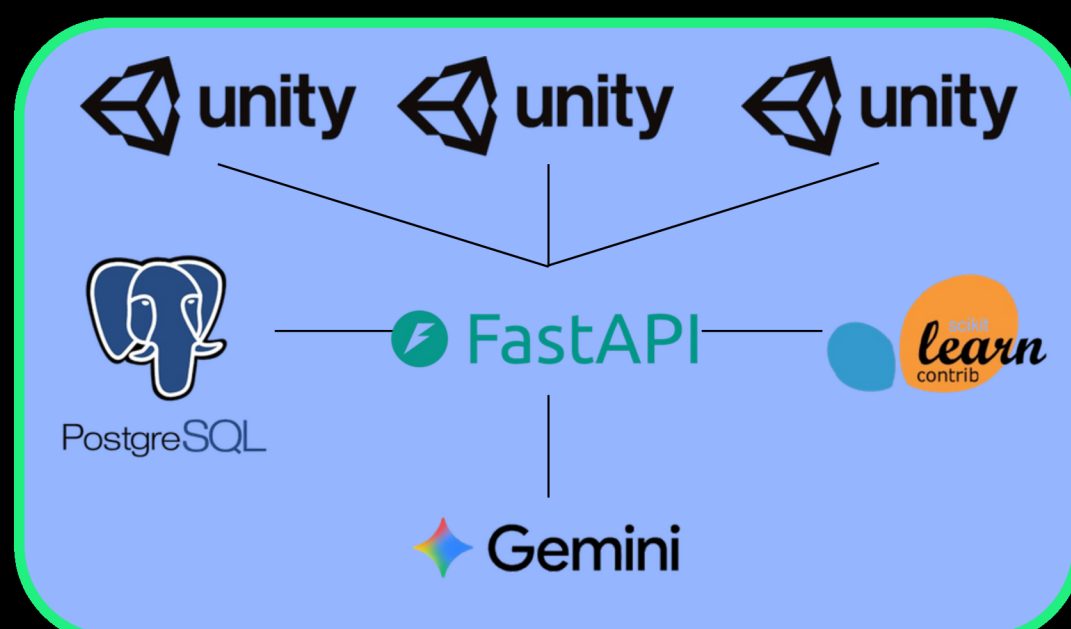
The player's objective is to **maximize overall city score** through **population growth** and **citizen satisfaction**. To achieve this, players must make strategic decisions by balancing resources, pollution, services, and expansion.

METHODOLOGY



- Agile-Inspired Workflow:** Adapted for solo development with flexible, iterative cycles
- Backlog & Iteration:** Tasks broken into small units, prioritized and refined weekly through playtesting
- Continuous Validation:** Focus on both functionality and "fun factor" with milestone-based progress

SYSTEM ARCHITECTURE



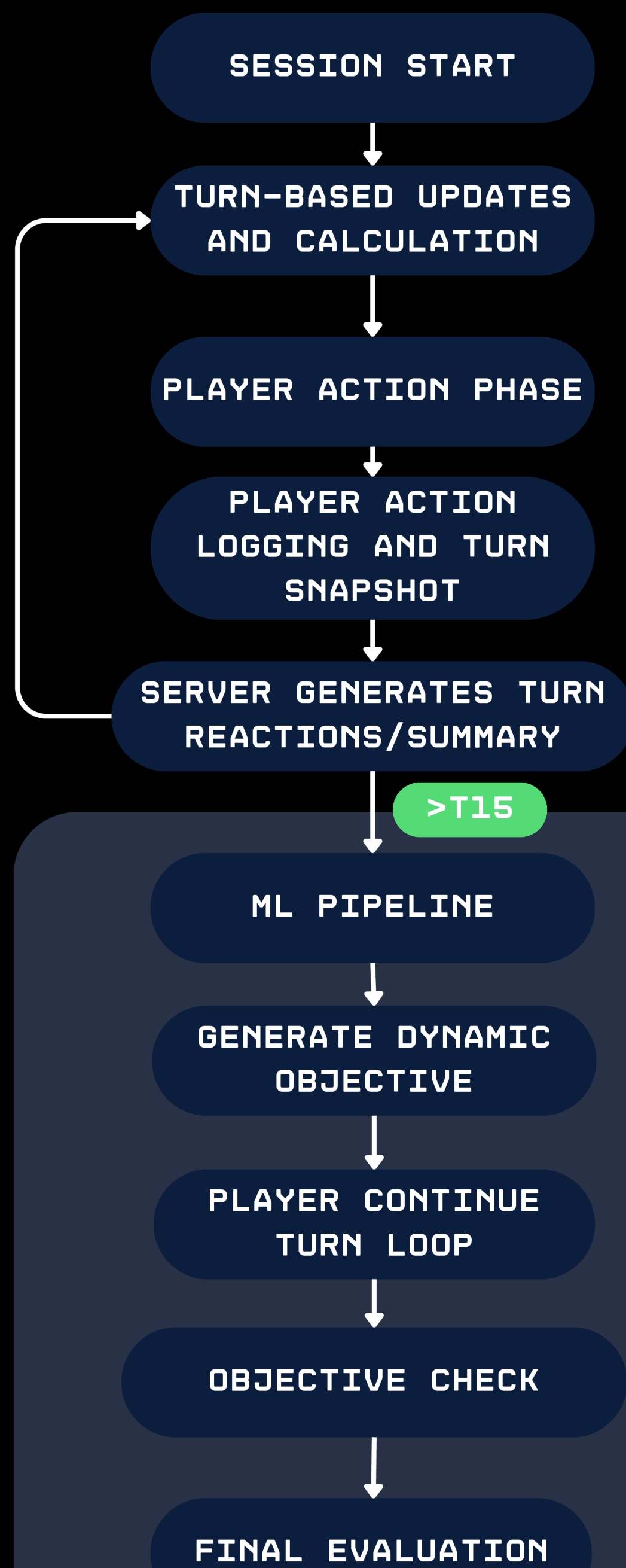
The system uses a **client-server architecture** where the Unity client handles game logic and validation, while the FastAPI server manages AI processing, data storage, and external API integration.

Both systems follow a **layered design**, with controllers routing requests to services that manage core game models and logic.

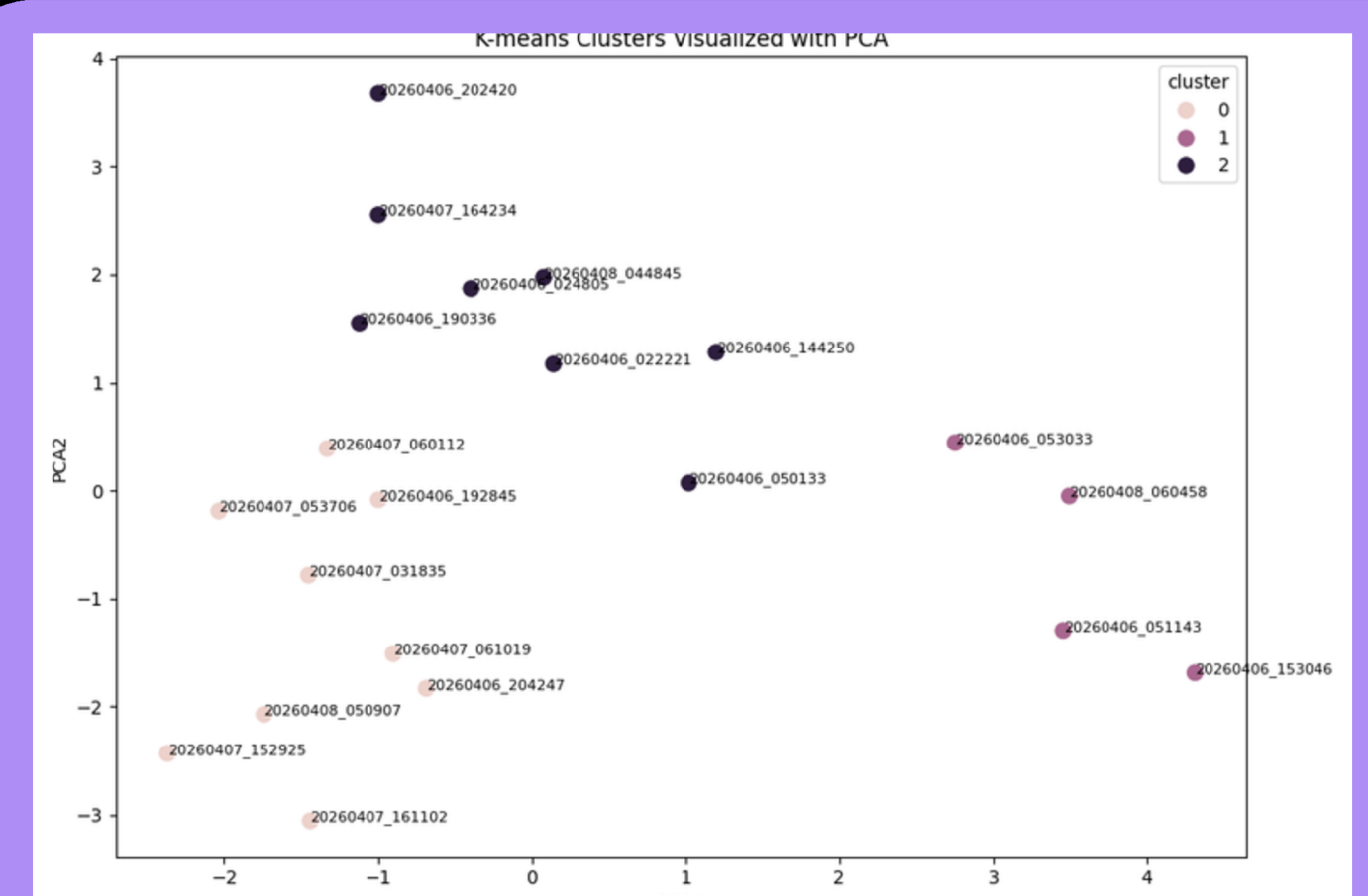
RESULTS AND DISCUSSIONS



GAME LOOP



ADAPTIVE
GAMEPLAY



THREE DISTINCT PLAYER CLUSTERS IDENTIFIED USING K-MEANS AND VISUALIZED WITH PCA, REPRESENTING DIFFERENT PLAYSTYLES SUCH AS EXPANDERS, BALANCED PLANNERS, AND SPECIALIZED STRATEGIES.

FEATURE EXTRACTION

PLAYSTYLE PREDICTION (CLUSTERING)

GROWTH PREDICTION (LINEAR REGRESSION)

LLM DECISION



```
{
  "playstyle": "Expander",
  "analysis": "High growth, low service coverage",
  "objective": "Reach 0.75 satisfaction by Turn 20",
  "reward_gold": 800,
  "difficulty": "Medium",
  "feedback": "Improve services to support your growing city."
}
```

EXAMPLE OF AN LLM-GENERATED RESPONSE
USED TO GUIDE PLAYER DECISIONS
THROUGH ADAPTIVE OBJECTIVES AND
FEEDBACK.

This game loop integrates an ML pipeline to analyze player behavior, predict playstyles and growth, and generate adaptive objectives and feedback to dynamically guide gameplay.

CONCLUSION

This study demonstrates that integrating an AI-driven pipeline in game design can effectively address limitations of rigid systems, repetitive gameplay, and lack of player guidance.

The results highlight AI's capability to deliver adaptive, engaging, and scalable gameplay experiences, supporting its potential in future game development.